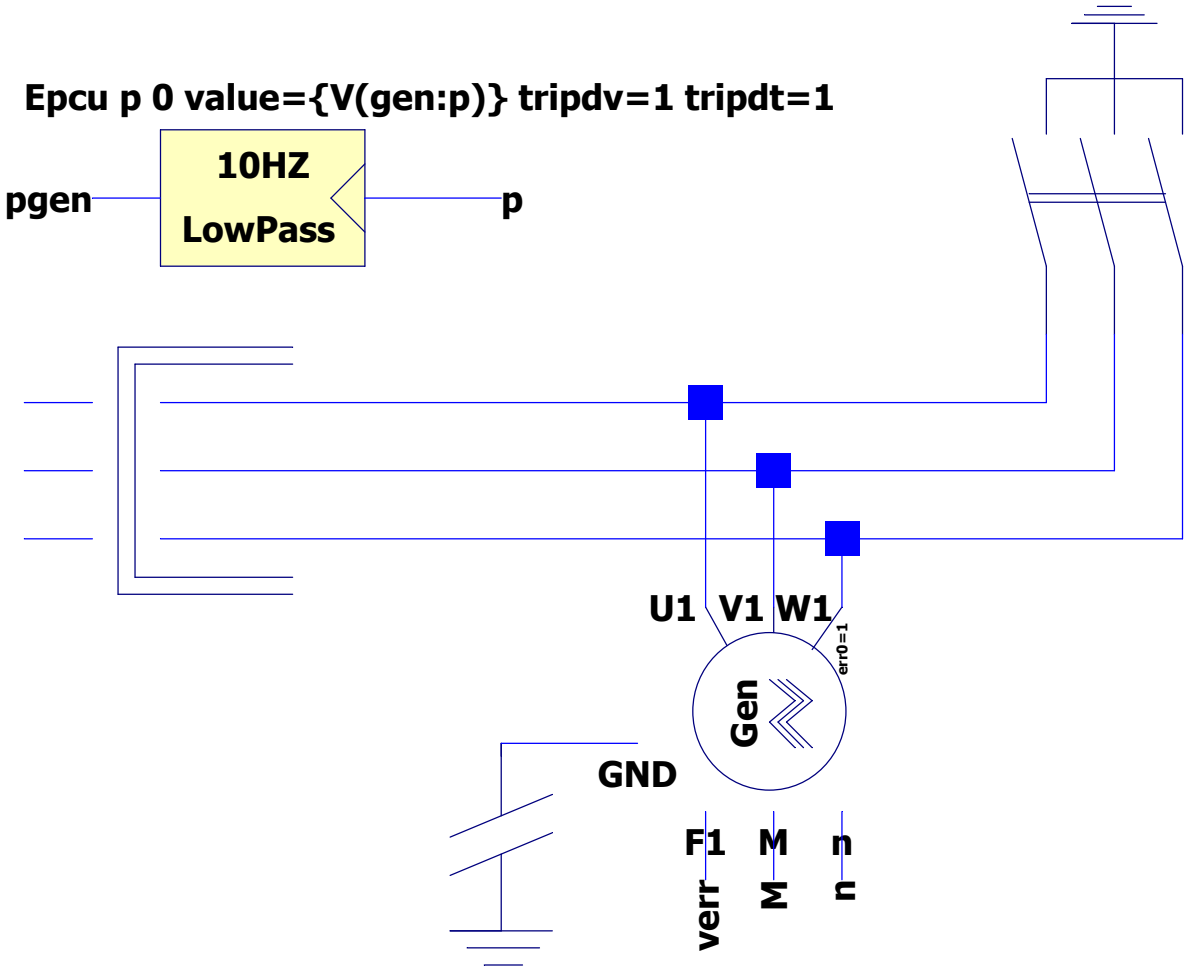
```
.tran 0 20s 11.5s 0.6m
.param tdv=1 tdt=1 fn1=60 omega_n={2*pi*fn1}
.options vntol=10 abstol=100m reltol=0.01
```

```
Epcu p 0 value={V(gen:p)} tripdv=1 tripdt=1
```




```
Vvr verr 0 200
```

```
*Subtransient: damping present
.meas TRAN USXd FIND V(gen:usx) AT 515.95ms
.meas TRAN USYd FIND V(gen:usy) AT 515.95ms
.meas TRAN ISXd FIND V(gen:isx) AT 515.95ms
.meas TRAN ISYd FIND V(gen:isy) AT 515.95ms
.meas TRAN IMXd FIND V(gen:imx) AT 515.95ms
.meas TRAN IMYd FIND V(gen:imy) AT 515.95ms
.meas TRAN IDXd FIND V(gen:idx) AT 515.95ms
.meas TRAN IDYd FIND V(gen:idy1) AT 515.95ms
* See output at VIEW spice output log
```

For phasor diagramm of first 15.95ms of shorting see also GeoGebra file: Synchrongenerator_with_damper.ggb and Wxmaxima file: Synchrongenerator_with_damper.wxmx